

Recap.

- ① H_0 : Default statement. H_1 : Opposite statement.
- ② Type I Error: H_0 holds but H_0 been rejected.
Type II Error: H_1 holds but H_0 not been rejected.
- ③ For H_0 : An unknown parameter $\mu = \mu_0$. $H_1 = \begin{cases} \mu \neq \mu_0 & \text{two-sided} \\ \mu > \mu_0 & \text{one-sided} \end{cases}$
- ④ Compare observed statistic value and theoretical value based on H_0 .
Model observed dataset $x_1, \dots, x_n$ as RVs $X_1, \dots, X_n$, compute $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.
(i) if population variance σ known, use z-test: $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$.
(ii) if σ unknown, only the variance of samples S_n^2 given, t-test: $T = \frac{\bar{X} - \mu_0}{S_n/\sqrt{n}} \sim t_{n-1}$.
Test statistic Z and T measures how far the sample data deviates from H_0 .
- ⑤ Significance level (α): the probability of type I Error happens, $P_{H_0}(\text{Reject } H_0)$.
(β : the probability of type II Error happens).
Rejection region: (i) for z-test with level $\alpha \in (0, 1)$, reject H_0 if $\begin{cases} |Z| > Z_{\alpha/2}, & \text{two-sided.} \\ Z > Z_\alpha, & \text{one-sided.} \end{cases}$
 Z_α is $1-\alpha$ -quantile of $N(0, 1)$.
(ii) for t-test with level α , reject H_0 if $\begin{cases} |T| > t_{n-1, \alpha/2} & \text{two-sided.} \\ T > t_{n-1, \alpha} & \text{one-sided.} \end{cases}$
- ⑥ Power ($1-\beta$): the probability of correctly reject H_0 when H_1 is true.
- ⑦ For $H_0: \theta = \theta_0$ and $H_1: \theta = \theta_1$, reject H_0 if $\frac{L(\theta_1|x)}{L(\theta_0|x)} > k$.

Exercises.

Ex 1. Hint: $Z_{0.025} \approx 1.96$, $Z_{0.0005} \approx 3.29$.

(i) $H_0: \mu = \mu_0 = 1200$. $H_1: \mu \neq 1200$.

(ii) σ known, use z-test, $Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{1250 - 1200}{150/\sqrt{36}} = 2$.

(iii) Reject H_0 if $|Z| > Z_{\alpha/2}$. When $\alpha = 0.05$, $|Z| = 2 > Z_{0.025}$, reject H_0 .

When $\alpha = 0.001$, $|Z| = 2 < Z_{0.0005}$, fail to reject H_0 .

Meaning: a smaller α requires stronger reason to reject the hypothesis.

Ex 2. Hint: For $X \sim \text{Exp}(\lambda)$, $P_X(x) = \lambda e^{-\lambda x}$ ($x \geq 0$).

$$(i) \Lambda(x) = \frac{L(\lambda_1|x)}{L(\lambda_0|x)} = \frac{\lambda_1 e^{-\lambda_1 x}}{\lambda_0 e^{-\lambda_0 x}} = \frac{\lambda_1}{\lambda_0} e^{(\lambda_0 - \lambda_1)x}$$

$$(ii) \Lambda(x) = \frac{\lambda_1}{\lambda_0} e^{(\lambda_0 - \lambda_1)x} > k',$$

$$e^{(\lambda_0 - \lambda_1)x} > \frac{k' \lambda_0}{\lambda_1},$$

$$(\lambda_0 - \lambda_1)x > \ln k' + \ln\left(\frac{\lambda_0}{\lambda_1}\right), \quad (\lambda_1 > \lambda_0, \lambda_0 - \lambda_1 < 0)$$

$$x < \frac{\ln k' + \ln\left(\frac{\lambda_0}{\lambda_1}\right)}{\lambda_0 - \lambda_1} = c.$$

$$(iii) \alpha = P_{H_0}(\text{Reject } H_0) = P_{H_0}(x < c) = \int_0^c \lambda_0 e^{-\lambda_0 x} dx = 1 - e^{-\lambda_0 c}, \quad c = -\frac{\ln(1-\alpha)}{\lambda_0}.$$

$$(iv) \beta = P_{H_1}(H_0 \text{ not rejected}) = P_{H_1}(x \geq c) = \int_c^\infty \lambda_1 e^{-\lambda_1 x} dx = e^{-\lambda_1 c}.$$

$$\text{Power of the test is } 1 - \beta = 1 - e^{-\lambda_1 c} = 1 - (1 - \alpha)^{\frac{\lambda_1}{\lambda_0}}.$$

Ex 3.

$$(i) H_0: \mu = 330. \quad H_1: \mu \neq 330.$$

(ii) σ unknown, use t-test, $T \sim t_{n-1} = t_9$, t-distribution with 9 degrees of freedom.

$$(iii) T = \frac{\bar{X}_n - \mu_0}{S_n / \sqrt{n}} = \frac{328.5 - 330}{\sqrt{4} / \sqrt{10}} \approx -2.373. \quad |T| = 2.373 > t_{9, 0.025} \approx 2.262, \text{ reject } H_0.$$

Ex 4. Hint: $\Phi(0.17) \approx 0.5675$.

$$(i) \sigma \text{ known, use Z-test with } \mu_0 = 10, \quad Z = \frac{\bar{X}_n - \mu_0}{\sigma / \sqrt{n}} = \frac{\bar{X}_{25} - 10}{3 / \sqrt{25}} = \frac{\bar{X}_{25} - 10}{0.6}.$$

Since this is one-sided test, reject H_0 if $Z > Z_{0.01} \approx 2.33$, i.e. $\bar{X}_{25} > 11.398$.

$$(ii) \beta = P_{H_1}(H_0 \text{ not rejected}) = P_{\mu = \mu_a = 11.5}(\bar{X}_{25} \leq 11.398) \quad (\text{use cdf to operate } \bar{X}_{25})$$

$$= P\left(\frac{\bar{X}_{25} - 11.5}{0.6} \leq \frac{11.398 - 11.5}{0.6}\right)$$

$$= P(Z_{25} \leq -0.17)$$

$$= 1 - \Phi(0.17) \approx 43.25\%.$$

$$(iii) \text{Power of the test is } 1 - \beta \approx 1 - 43.25\% = 56.75\%.$$